

Title

Neurobiological mechanisms in psychological treatment of anxiety: protocol for a randomized trial examining neurotrophic factor changes during acceptance-based versus supportive psychotherapy

Abstract

Background: Psychological interventions produce clinical improvement in anxiety disorders, but the underlying neurobiological mechanisms are incompletely understood. Neurotrophic factors regulate neural plasticity and may represent a biomarker of therapeutic response.

Methods: This 14-week randomized trial will enroll 85 adults aged 18-65 with diagnosed generalized anxiety disorder. Participants will be randomized 1:1 to either group acceptance-based behavior therapy (ABBT) or supportive therapy (ST), each delivered across 10 sessions. Blood samples for plasma neurotrophic factor measurement will be collected at baseline and post-treatment. Clinical anxiety severity will be assessed using standardized rating scales. Participants will complete self-report measures of worry, anxiety symptoms, and functional impairment. Correlations between neurotrophic factor changes and clinical outcomes will be examined using Pearson correlation and linear regression analyses.